the capsule transmits the result on a laser beam up to the ship: “299,792 kilometers per second; 299,792; 299,792; 299,792 . . .” The color of the incoming laser beam shifts from green to red to infrared to microwave to radio as the capsule nears the horizon, and still the message is the same: “299,792; 299,792; 299,792 . ..” And then the laser beam is gone. The capsule has pierced the horizon, and never once as it fell was there any change in the speed of light inside it, nor was there any change in the laws of physics that governed the workings of the capsule’s electronic systems.

These experimental results please you greatly. In the early twentieth century Albert Einstein proclaimed, largely on philosophical grounds, that the local laws of physics (the laws in regions small enough that one can ignore the curvature of spacetime) should be the same everywhere in the Universe. This proclamation has been enshrined as a fundamental principle of physics, the *equivalence principle*.[32] Often in the ensuing centuries the equivalence principle was subjected to experimental test, but never was it tested so graphically and thoroughly as in your experiments here near Gargantua’s horizon.

You and your crew are now tiring of the struggle with 10 Earth gravities, so you prepare for the next and final leg of your voyage, a return to our Milky Way galaxy. Your crew will transmit an account of your Gargantua explorations during the early stages of the voyage; and since your starship itself will soon be traveling at nearly the speed of light,; the transmissions will reach the Milky Way less than a year before the ship, as measured from Earth.

As your starship pulls up and away from Gargantua, your crew make a careful, telescopic study of the quasar 3C273 overhead[33] (Figure P.5).

Its jets—thin spikes of hot gas shooting out of the quasar’s core—are enormous: 3 million light-years in length. Training your telescopes on the core, your crew see the source of the jets’ power: a thick, hot, doughnut of gas less than 1 light-year in size, with a black hole at its center. The doughnut, which astrophysicists have called an “accretion disk,” orbits around and around the black hole. By measuring its rotation period and circumference, your crew infer the mass of the hole: 2 billion (2×10^9) solar masses, 7500 times smaller than Gargantua, but far larger than any hole in the Milky Way. A stream of gas flows from the doughnut to the horizon, pulled by the hole’s gravity. As it nears the horizon the stream,